

Title	 Diamond Dynamics: Price Prediction and Market Segmentation
Skills take away from the project	• Data Cleaning & Preprocessing • EDA & Data Visualization • Feature Engineering • Outlier & Skewness Handling • Regression using ML algorithms, including ANN • K-Means Clustering & Cluster Labeling • Dimensionality Reduction with PCA • Streamlit UI Design
Domain	E-Commerce , Luxury Goods Analytics , Retail Pricing Optimization

Problem Statement:

The diamond market heavily depends on multiple quality attributes like carat, cut, clarity, and color to decide pricing. Accurately predicting diamond prices helps in pricing strategy and customer targeting. Additionally, segmenting diamonds into meaningful market groups can assist sellers in inventory management and buyers in finding suitable products.

Objectives:

- Predict diamond prices using different regression models and Artificial Neural Networks (ANN).

- Cluster diamonds into market segments based on physical and qualitative features.
 - Develop an interactive Streamlit UI for price prediction and cluster identification based on user inputs.
-

Real-World Use Cases:

- **Dynamic pricing strategy** for diamond retailers.
 - **Inventory categorization** and market segmentation for product listing.
 - **Luxury product recommendation** engines based on diamond profiles.
 - **Customer segmentation** and personalized marketing.
-

Data Collection:

- **Source:** [Diamond Dataset](#)
 - **Shape :** 53940 rows and 10 features
-

Dataset — Column Descriptions

Column Name	Description
carat	Weight of the diamond (in carats). A primary factor influencing price.
cut	Quality of the diamond's cut. Ordinal categories: <i>Fair, Good, Very Good, Premium, Ideal</i> .

color	Diamond color grading from best (D) to worst (J). Ordinal categories: <i>D, E, F, G, H, I, J</i> .
clarity	Measure of diamond inclusions (internal flaws) and blemishes. Categories: <i>IF (Internally Flawless), VVS1, VVS2, VS1, VS2, SI1, SI2, I1</i> .
depth	Total depth percentage = $(z / \text{mean}(x, y)) \times 100$. Affects brilliance and appearance.
table	Width of the diamond's top facet as a percentage of the average diameter. Impacts how light reflects internally.
price	Price of the diamond in USD. Target variable(Convert to INR)
x	Length of the diamond (in mm).
y	Width of the diamond (in mm).
z	Depth (height) of the diamond (in mm).

Data Preprocessing:

- Handle missing values (if any)
 - Convert zero or invalid values in **x**, **y**, **z** to null or imputed values
 - Drop unnecessary columns if irrelevant for prediction/clustering
-

Outliers Handling:

- **IQR Method / Z-Score Method** for numerical features (`carat`, `price`, `x`, `y`, `z`)
 - Visual checks via boxplots
-

Skewness Handling:

- Check skewness using `.skew()` function
 - Apply **log transformation** / **square root** / **box-cox** on highly skewed variables like `price`, `carat`, `x`, `y`, `z`
-

Exploratory Data Analysis (EDA):

Analyses & Visualizations ideas:

- Distribution plots for `price`, `carat`, `x`, `y`, `z`
- Count plots for categorical features (`cut`, `color`, `clarity`)
- Price variation with `carat`, `cut`, `color`, `clarity` using boxplots
- Correlation heatmap for numerical features
- Scatterplot matrix for `carat`, `x`, `y`, `z`, and `price`
- Pairwise relationships using `sns.pairplot()`

- Carat vs. price regression lineplot
 - Average price per **cut**, **color**, **clarity** categories using bar plots
-

Feature Engineering:

Convert the price (in USD) to INR using a fixed or dynamic conversion rate

New Columns that can be Derived:

- Volume = $x * y * z$
- Price per Carat = price / carat
- Dimension Ratio = $(x + y) / (2 * z)$
- Carat Category: Light (<0.5), Medium ($0.5-1.5$), Heavy (>1.5)



Note:

 Learners can engineer as many additional meaningful features by exploring relationships between **carat**, **price**, **cut**, **color**, **clarity**, dimensions (**x**, **y**, **z**), or combining categorical and numerical variables. Creative, domain-relevant feature engineering often improves model performance.

Feature Selection

To improve model performance and reduce redundancy, you can select the most relevant features using any **one of the following techniques**:

Available Techniques:

- **Correlation Matrix Analysis** — To detect multicollinearity among numerical features.
- **Feature Importance** — Using models like Random Forest or XGBoost to rank features by importance.
- **Recursive Feature Elimination (RFE)** — Iteratively removes least important features.
- **Variance Inflation Factor (VIF)** — To detect multicollinearity among numerical features.

👉 **Note:**

You may choose and apply any one of the above feature selection techniques based on your preference and model requirements.

📌 **Encoding**

Perform encoding of the categorical columns based on the requirements of your models. You can use techniques like **Label Encoding**, **Ordinal Encoding**, or **One-Hot Encoding** depending on whether the categorical variables are ordinal or nominal, and based on what the specific algorithm requires.

📌 **Model Building:**

📌 **1 Regression (Price Prediction)**

- Split data into Train-Test sets (80-20 or 70-30).

- Build 4–5 ML regression models (e.g., Linear Regression, Random Forest, XGBoost, Decision Tree, KNN).
 - Build an **ANN** regression model.
 - Evaluate all models using MAE, MSE, RMSE, R^2 .
 - Compare model performances and save the best performing model as .pkl file
-

2 Clustering (Market Segmentation)

- Try K-Means clustering and **optionally** try different clustering techniques (e.g., **K-Means, DBSCAN, Hierarchical Clustering**).
 - Use the **Elbow Method** or **Silhouette Score** to find the optimal number of clusters.
 - **Encode categorical variables** (**cut**, **color**, **clarity**) before fitting the model.
 - Optionally, perform **Dimensionality Reduction with PCA**:
 - Reduce features to 2 or 3 principal components.
 - Use PCA for **visualizing clusters in a 2D or 3D scatter plot**.
 - Pickle the **best-performing clustering model(pkl)** for use in Streamlit.
-

Cluster Naming Approach:

After clustering, analyze average **price**, **carat**, and **cut** distribution per cluster.

- Name clusters based on characteristics like:

- "Premium Heavy Diamonds"
- "Affordable Small Diamonds"
- "Mid-range Balanced Diamonds"

Cluster Naming Ideas Based on Characteristics

Cluster Characteristic	Example Cluster Name	Description
High carat, high price	Premium Heavy Diamonds	Large, expensive, premium-grade stones
Low carat, low price	Affordable Small Diamonds	Small, budget-friendly stones
Medium carat, medium price	Mid-range Diamonds	Balanced in size and cost

Streamlit App Features

1 Price Prediction Module

Objective:

Predict the price of a diamond based on its attributes using the trained regression model.

Functionality example:

- Numeric inputs for: `carat, x, y, z`
- Select box inputs for: `cut, color, clarity`
- Button: **Predict Price**
- Display: Predicted diamond price in INR

2 Market Segment Prediction (Clustering Module)

Objective:

Predict the cluster (market category) the diamond belongs to, using the trained clustering model.

Functionality:

- Use the same input form as price prediction
- Button: **Predict Cluster**
- Display: Cluster number and mapped cluster name (e.g., *Premium Heavy Diamonds*, *Affordable Small Diamonds*)



Project Deliverables

Python Notebook with:

- Clean, well-documented code with comments
- Visualizations for EDA and clustering insights
- Feature engineering and feature selection
- 4–5 **ML regression** models + **ANN** model for price prediction(Use tf or pytorch)
- Clustering model (dropping **price**, using relevant features)
- Model evaluation metrics (MAE, MSE, RMSE, R^2 for regression, inertia/silhouette score for clustering)

Streamlit Web Application:

- User input for diamond attributes
 - Predicts price using trained regression model
 - Predicts market cluster segment using trained clustering model
 - Displays cluster name and visual insights
 - Clean, interactive UI with real-time outputs
-

Technical Tags

machine learning, regression, artificial neural network, random forest, decision tree, xgboost, knn, model evaluation, feature engineering, feature selection, feature importance, outlier detection, skewness treatment, clustering, k-means, elbow method, silhouette score, streamlit, model deployment, diamond price prediction, market segmentation, scikit-learn, tensorflow, keras, exploratory data analysis, eda, matplotlib, seaborn, pickle model saving, numeric input handling, web app development

Timeline

The complete project must be finished and submitted within **8 days** from the assigned date.

References

Streamlit recording (English)	 Special session for STREAMLIT(11/08/2024)
Streamlit Reference doc	Streamlit API reference
Project Live Evaluation	 Project Live Evaluation
Capstone Explanation Guideline	 Capstone Explanation Guideline
GitHub Reference	 How to Use GitHub.pptx
Machine Learning(Eng) Recommendation systems	 Project Excellence Series: Guided Lear...
Machine Learning(Tam) Recommendation systems	 Project Excellence Series: Guided Lear...
Unsupervised learning study material	  Unsupervised Learning
Deep Learning Study material	 Deep_Learning-study_material.pdf
Project Orientation (Both Eng & Tam)	 Diamond Dynamics: Price Prediction an...

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: You can join the doubt session using the link provided in sheet given below

Timing: Monday-Friday(3.00PM to 4.00PM), Saturday (2.00PM to 3.00PM)

Booking link : [Project Doubt Session \[DS/AI/ML\]](#)

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday(11.00 am to 12:00 pm) (05:00PM to 06:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

Created By	Verified By	Approved By
Nilofer Mubeen	Shadiya	Nehlath Harmain